Homework 8

Start Assignment

- Due Tuesday by 3pm
- Points 20
- Submitting a website url
- Available Oct 20 at 5:30pm - Oct 30 at 11:59pm

Assignment: Mini Diffusion Model (DDPM → DDIM)

In this assignment, you will implement a simplified diffusion model inspired by the DDPM paper and modify its sampling procedure to reproduce DDIM sampling. The goal is to understand how diffusion models generate images by adding noise to data and learning to reverse this process. Use the MNIST dataset and train a small CNN noise-prediction network.

Configuration:

- Diffusion steps: T = 100
- Noise schedule: β_t linearly increasing from 0.0001 to 0.02
- Recommended training: batch size 128, epochs 50.

Tasks

(a) Forward Diffusion (5 pts)
Implement the forward diffusion process

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\epsilon$$

where $\epsilon \sim N(0, 1)$).
Visualize one MNIST image at several timesteps (e.g., $t = 0, 10, 30, 60, 100$).

(b) Train a Mini DDPM (7 pts)
Train a CNN (that contain **at least 4 convolutional layers**, use **nonlinear activations (e.g., ReLU)**, **condition on the timestep t** and output a tensor with the **same shape as the input image)** to predict the noise added to an image using the loss

$$L = MSE(\epsilon, \epsilon_\theta(x_t, t))$$

Plot the training loss.

(c) DDPM Sampling (4 pts)
Starting from Gaussian noise, implement the **DDPM reverse sampling process** to generate images.

(d) DDIM Sampling (4 pts)

Modify the sampling procedure to implement DDIM sampling.

Generate images using:

- DDPM (100 steps)
- DDIM (50 steps)
- DDIM (20 steps)

Compare the outputs.

Deliverables

Submit a notebook containing:

- forward diffusion visualization
- training loss plot
- generated images (DDPM vs DDIM)
- discussion of the differences.